

Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

Executive Summary

This project presents a comprehensive analysis of the **Factory-to-Customer Shipping Route Efficiency for Nassau Candy Distributor**. The primary objective was to evaluate shipping performance, identify operational bottlenecks, compare shipping modes, and develop an interactive business intelligence dashboard to support data-driven logistics decisions.

The analysis was performed on **10,194 shipment records** using Python, Pandas, Plotly, and Streamlit. Key performance indicators, route-level analytics, regional performance, congestion analysis, and transportation cost efficiency were evaluated to identify opportunities for improving supply chain operations.

Business Objectives

The project aimed to answer the following business questions:

- Which factory-to-customer routes operate most efficiently?
 - Which regions experience operational bottlenecks?
 - Which shipping mode provides the best balance between delivery performance and transportation cost?
 - How does shipment volume influence delivery performance?
 - How can logistics managers monitor shipment performance through an interactive dashboard?
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Executive Highlights

KPI	Result
Total Shipments	10,194
Average Transportation Cost	4.74
Average Profit	9.17
Shipping Modes Evaluated	4
Geographic Regions Analyzed	4
States Evaluated	50+

Note: The original dataset contains **Order-to-Shipment Duration** values averaging approximately **1,320.84 days**, calculated directly from the provided Order Date and Ship Date fields. These values were preserved from the source data and displayed in the dashboard.

Key Findings

1. Regional Performance

Regional analysis revealed measurable differences in shipment performance.

Region	Avg. Lead Time (Days)	Total Shipments	Category
Atlantic	4.510	2,986	Critical Bottleneck
Gulf	4.461	1,620	Efficient
Interior	4.531	2,335	Slow but Low Volume
Pacific	4.481	3,253	Efficient

Business Insight

- Atlantic requires immediate operational attention because it combines relatively high lead time with substantial shipment volume.
 - Pacific successfully manages the highest shipment volume while maintaining efficient delivery performance.
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2. Shipping Mode Performance

Four transportation methods were evaluated.

Ship Mode	Avg. Lead Time	Avg. Cost	Shipments
First Class	4.461	4.721	1,548
Same Day	4.484	4.411	547
Second Class	4.502	4.828	1,979
Standard Class	4.507	4.749	6,120

Business Insight

- **Standard Class** is the organization's primary shipping method, accounting for approximately **60%** of all shipments.
 - **First Class** provides the shortest average delivery time.
 - **Same Day** records the lowest average transportation cost.
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3. Efficiency versus Cost

Transportation efficiency was evaluated using the Efficiency per Cost ratio.

Ship Mode	Efficiency per Cost
Same Day	1.017
Standard Class	0.949
First Class	0.945
Second Class	0.933

Business Insight

Same Day shipping demonstrates the most favorable balance between transportation cost and delivery performance, while Second Class provides the lowest efficiency relative to cost.

4. Regional Bottlenecks

The bottleneck analysis identified operational priorities across the distribution network.

Region	Bottleneck Score
Pacific	14,577
Atlantic	13,467
Interior	10,581
Gulf	7,227

Although the Pacific region recorded the highest bottleneck score because of its shipment volume, its delivery performance remained efficient. Atlantic represents the highest operational priority because it combines relatively high shipment volume with above-average lead time.

5. Interactive Dashboard

A fully interactive dashboard was developed using **Streamlit** and **Plotly** to support operational decision-making.

The dashboard enables users to:

- Monitor logistics KPIs
- Analyze factory-to-customer routes
- Compare shipping modes
- Visualize geographic shipment performance
- Identify regional bottlenecks
- Drill down to state-level shipment data
- Filter results dynamically by date, region, state, and shipping mode

These capabilities are demonstrated throughout the uploaded dashboard.

Business Value

The project provides measurable business value by:

- Improving visibility into logistics performance.
- Identifying regions requiring operational improvements.
- Supporting evidence-based transportation planning.
- Enabling managers to compare shipping modes using performance and cost metrics.
- Providing an interactive platform for monitoring shipment operations in real time.

Recommendations

Based on the findings, the following actions are recommended:

1. Prioritize operational improvements in the Atlantic region to reduce delivery delays.
2. Continue leveraging Standard Class for high-volume shipments while monitoring service performance.
3. Investigate why Second Class incurs the highest average transportation cost without delivering proportionate improvements in delivery speed.
4. Use Same Day shipping selectively where its favourable cost-performance balance aligns with customer and operational requirements.
5. Expand the dashboard with predictive analytics to forecast delivery delays and optimize route planning.

Conclusion

The Factory-to-Customer Shipping Route Efficiency Analysis successfully transformed shipment records into actionable business intelligence. Through comprehensive analysis of **10,194 shipments**, the project identified regional bottlenecks, evaluated shipping mode performance, assessed transportation cost efficiency, and delivered an interactive dashboard for operational monitoring. These insights provide management with a practical foundation for improving logistics efficiency, optimizing transportation strategies, and enhancing customer service through data-driven decision-making.
